

AMERICAN TERAWATT

HVDC Configurations

System Topologies

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Components

Terminal HV components overview

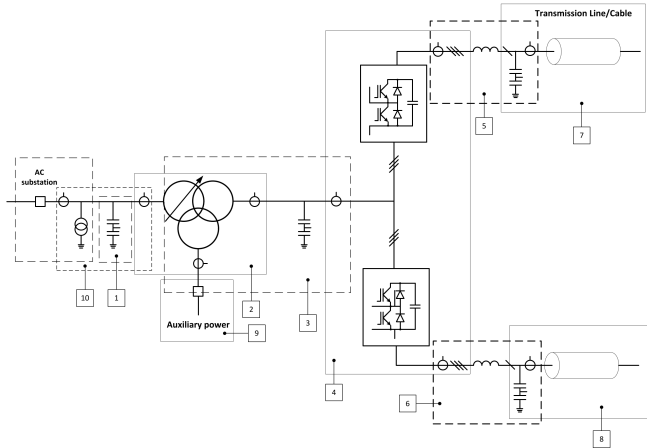
HV components overview

- Converter
- Transformer
- AC Reactor
- DC Reactor
- Overvoltage protection
- Pre-Insertion Resistor (PIR)
- Star point Reactor (SPR)
- Wall Bushings
- Earthing Devices
- Disconnectors
- AC Filter
- DC Filter
- RI Filter
- PLC Filter
- Measuring Transducers
- Valve Cooling System
- Auxiliary Power Supply
- Circuit Breakers

Measuring scheme

Voltage and current transducers

Overview



- Current measurement
- Voltage measurement

CIGRÉ 832 - Guide for Electromagnetic Transient Studies
Involving VSC Converters

Measuring transducers

- Measure the analog voltage and current signals required for monitoring and control of the HVDC system.
- Located in the HV plant itself; usually convert the measured signal to a level and form suitable for subsequent processing ($< 5\text{ A}$, $< 10\text{ V}$).
- Any major scaling of the measured signal is implemented on the measurement device itself or on additional associated equipment.
- Provide galvanic (electrical) isolation between the raw HV signals and the clean environment of the control cubicles.

AC voltage measurement

AC voltage measurement can be done by:

- Inductive Voltage Transformer
- Capacitor Voltage Transformer (CVT)

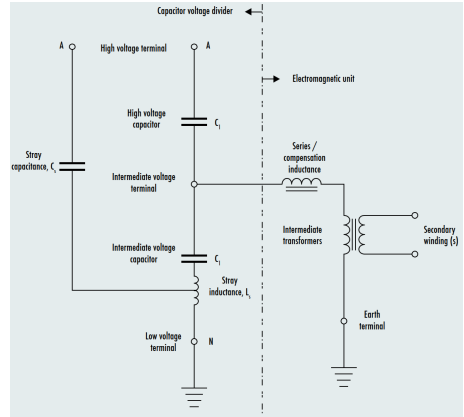


Fig. 6.6.3a -Typical line diagram of a capacitor voltage transformer. © GE Vernova, 2023.

AC current measurement

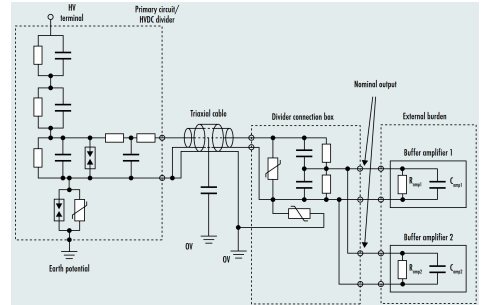
AC current transformer

- Measuring principle usable for AC current measurements only.
- Different insulators can be used: SF₆ or “Clean Air”.
- Lacks the capability to measure currents at low frequencies (< 5 Hz).



DC voltage measurement

Direct voltage on the HVDC busbar is measured by a DC resistive Voltage Divider (VD).



DC Transmission Systems: Line
Commutated Converters. © GE Vernova,
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DC current measurement

DC Current Transducer (DCCT) used for DC current measurement:

- Zero Flux Current Transformers (ZFCT) or Optical Current Transformers (OCT)
- Measurement accuracy of 0.2% or better
- Wide frequency range for very fast protection schemes
- ZFCT installation over a bushing or cable

